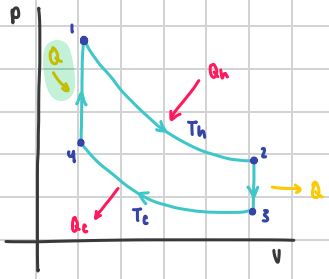
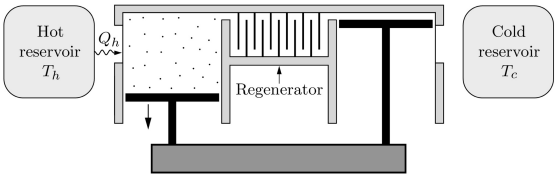
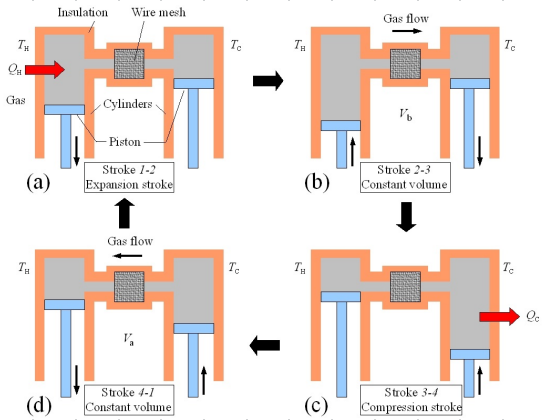
Carnot efficiency: $e = \frac{T_h - T_c}{T_h} = 1 - \frac{T_c}{T_h}$

STIRLING CYCLE







1→2: isothermal expansion (hot piston out, cold piston at rest inward)

2→3: isochoric cooling (hot piston in, cold piston out) → move part regenerator gives up heat, cool to T_c

3→4: isothermal compression (cold piston in, hot piston at rest inward)

4→1: isochoric heating (cold piston in, hot piston out) → move part regenerator absorbs heat to T_h

Without regenerator, from 4→1, heat is absorbed from outside ⇒ lower efficiency

$$\eta = \frac{\left(1 - \frac{T_c}{T_h}\right) k \ln\left(\frac{V_2}{V_1}\right)}{k \ln \frac{V_2}{V_1} + c_v\left(1 - \frac{T_c}{T_h}\right)} < 1 - \frac{T_c}{T_h}$$

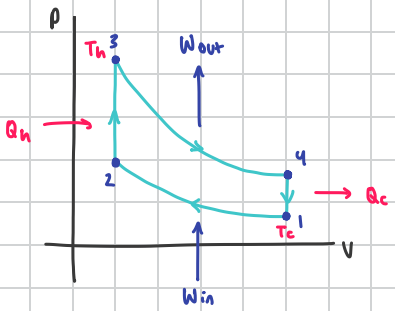
With regenerator, theoretically matches Carnot efficiency ($Q_{in} = Q_h$)

$$\eta = 1 - \frac{T_c}{T_h}$$

(ideal Sterling cycle)

gasoline cars

OTTO CYCLE



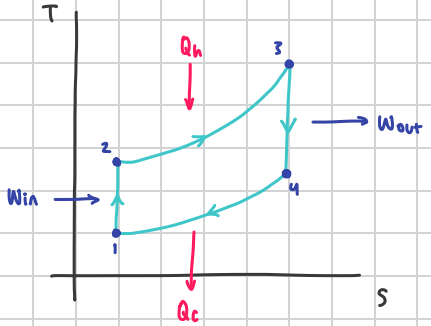
pV^γ = const.

1→2. Isentropic* compression (adiabatic)

2→3: Const. vol. heat addition

3→4: Isentropic* expansion (adiabatic)

4→1: Const. vol. heat ejection



$$e = 1 - \left(\frac{V_2}{V_1}\right)^{\gamma - 1}$$

γ: adiabatic exponent

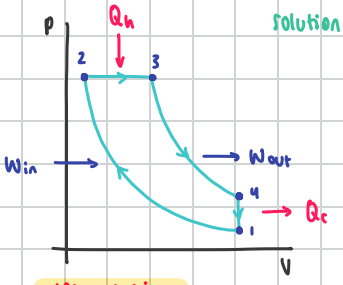
$\frac{V_2}{V_1}$: compression ratio

diesel trucks

DIESEL CYCLE

if fuel mixture too hot during compression, it may ignite before compression complete

solution: start by compressing pure air only, then inject fuel



1→2: Isentropic compression of pure air (adiabatic)

2→3: Isobaric expansion (fuel injected and combusts)

3→4: Isentropic expansion of hot gas (adiabatic)

4→1: Isochoric cooling

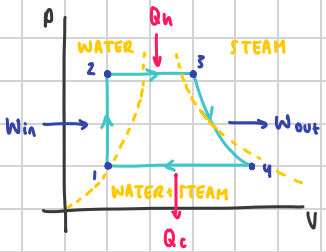
$$e = 1 - \left(\frac{V_2}{V_1}\right)^{\gamma - 1}\left[\frac{\alpha^\gamma - 1}{\gamma(\alpha - 1)}\right]$$

α is cutoff ratio $\frac{V_3}{V_2}$ (how long fuel is injected)

↑ for same compression ratio, Otto more efficient, but in practice diesel has higher ratio so more efficient

steam engine

RANKINE CYCLE



1→2 pump: increase pressure of liquid

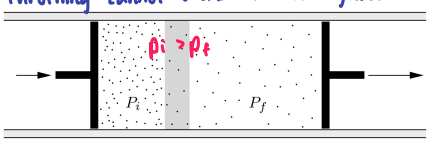
2→3 boiler: isobaric heating, liquid → superheated vapor $Q_{in} = H_3 - H_2$

3→4 turbine: adiabatic expansion of steam to do work

4→1 condenser: isobaric heat rejection, vapor → liquid

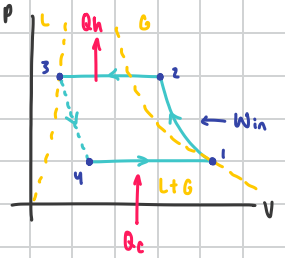
$$e = \frac{(H_3 - H_4) - (H_2 - H_1)}{H_3 - H_2} = 1 - \frac{H_4 - H_1}{H_3 - H_2}$$

Throttling cannot work for ideal gases!



REAL REFRIGERATORS

reverting the Rankine cycle



1→2 compressor: adiabatic compression

2→3 condenser: isobaric heat rejection $Q_{expelled} = H_2 - H_3$

3→4 throttle: adiabatic, no work, isenthalpic, fluid forced through narrow restriction into lower pressure

4→1 evaporator: isobaric heating $Q_{absorbed} = H_1 - H_4$

$$COP = \frac{Q_{absorbed}}{Q_{expelled} - Q_{absorbed}} = \frac{H_1 - H_4}{H_2 - H_3 - H_1 + H_4} \approx \frac{H_1 - H_3}{H_2 - H_1}$$

$W = Q_{expelled} - Q_{absorbed}$

$H_3 \approx H_4$

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